Therefore, $\frac{3+4i}{5-12i} = -\frac{33}{169}+i\frac{56}{169}$. This is in the $x+iy$ form.

Hence real part is $-\frac{33}{169}$ and imaginary part is $\frac{56}{169}$.

Example 2.4

Simplify $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$. into rectangular form

Solution

We consider $\frac{1+i}{1-i} = \frac{(1+i)(1+i)}{(1-i)(1+i)} = \frac{1+2i-1}{1+1} = \frac{2i}{2} = i$,

and $\frac{1-i}{1+i} = \left(\frac{1+i}{1-i}\right)^{-1} = \frac{1}{i} = -i$.

Therefore, $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1-i}\right)^3 = i^3 - (-i)^3 = -i - i = -2i$.

Example 2.5

If $\frac{z+3}{z-5i} = \frac{1+4i}{2}$, find the complex number z in the rectangular form

Solution

We have $\frac{z+3}{z-5i} = \frac{1+4i}{2}$

$\Rightarrow 2(z+3) = (1+4i)(z-5i)$

$\Rightarrow 2z+6 = (1+4i)z + 20 - 5i$

$\Rightarrow (2-1-4i)z = 20-5i-6$

$\Rightarrow z = \frac{14-5i}{1-4i} = \frac{(14-5i)(1+4i)}{(1-4i)(1+4i)} = \frac{34+51i}{17} = 2+3i$.

Example 2.6

If $z_1 = 3-2i$ and $z_2 = 6+4i$, find $\frac{z_1}{z_2}$ in the rectangular form

Solution

Using the given value for z_1 and z_2 the value of $\frac{z_1}{z_2} = \frac{3-2i}{6+4i} = \frac{3-2i}{6+4i} \times \frac{6-4i}{6-4i}$

$= \frac{(18-8)+i(-12-12)}{6^2+4^2} = \frac{10-24i}{52} = \frac{10}{52} - \frac{24i}{52}$

$= \frac{5}{26} - \frac{6}{13}i$.

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